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PROJECT

PROPOSALFORM

Project Title:	High-Speed Industrial "Black Box" Design for Vibration Analysis and Fault Diagnosis of Asynchronous Motors using STM32 Architecture
Thematic Area:	Embedded Systems Digital Signal Processing (DSP) Predictive Maintenance (PdM) Data Acquisition (DAQ)
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1. Abstract

This project aims to design and implement a high-fidelity "Industrial Black Box" for the condition monitoring and fault diagnosis of three-phase asynchronous induction motors using the STM32F407G-DISC1 (Discovery) platform. Induction motors are essential prime movers in modern industry; however, they are susceptible to mechanical degradations such as bearing wear, shaft imbalance, and axial misalignment, which often culminate in catastrophic failures and unscheduled downtime [1]. Traditional low-frequency monitoring systems often fail to capture the high-frequency acoustic emissions (10 kHz to 15 kHz) that occur during the incipient stages of bearing failure [2].

To bridge this gap, this study proposes a specialized sensing strategy utilizing three independent single-axis Piezoelectric transducers instead of traditional MEMS-based modules. This approach provides a dual advantage: superior bandwidth capability for capturing high-frequency "micro-screams" and extreme cost-effectiveness, making it a viable alternative to expensive industrial-grade vibration analyzers. Since these sensors generate high-impedance, low-level signals, a custom Analog Front-End (AFE) - including buffer and amplifier circuits - is designed to condition and scale the signals for the STM32 ADC.

To ensure absolute signal integrity at high sampling rates, the firmware architecture utilizes the ARM Cortex-M4 core's DMA engine with a Ping-Pong buffering strategy. Raw data is retrieved gaplessly and logged into an SD card via the high-speed SDIO (4-bit) protocol. The analytical core transitions from simple static spectrums to dynamic Short-Time Fourier Transform (STFT) based Spectrograms. This multi-domain approach allows for the identification of fault signatures even during motor transients (start/stop), providing a unique "visual fingerprint" for mechanical defects . The resulting system serves as a portable, high-precision, and affordable forensic tool that facilitates data-driven Predictive Maintenance in various industrial settings [3], [4].

2. Motivation and Objectives

2.1. Motivation

Three-phase induction motors are the primary driving force in contemporary industrial processes, accounting for nearly 90% of the energy consumed by industrial prime movers [1]. Any unplanned failure in these machines results in significant economic losses due to production downtime and the high cost of emergency repairs. Historically, maintenance has been either reactive (corrective) or preventive (time-based); however, both approaches are inefficient. Reactive maintenance risks catastrophic failure, while preventive maintenance often results in the premature replacement of healthy components [4].

The fundamental motivation of this project is to shift the maintenance paradigm toward Predictive Maintenance (PdM) by developing an "Industrial Black Box" for real-time condition monitoring. Mechanical faults such as rotor imbalance and bearing wear manifest as vibration signatures long before visible failure occurs [2]. While standard monitoring systems utilize low-frequency MEMS sensors, they often miss the high-frequency acoustic emissions (10 kHz+) typical of incipient bearing cracks. This project utilizes three independent single-axis piezoelectric transducers, which offer two critical advantages: a superior bandwidth of up to 15 kHz and an exceptionally low cost compared to professional industrial DAQs. By designing a custom Analog Front-End (AFE) to match the high-impedance nature of these sensors [7], we aim to capture these high-fidelity "micro-vibrations," providing a reliable forensic record that complies with ISO 20816 vibration standards [8].

2.2. Objectives

The primary objectives of this project are:

- **Cost-Effective High-Bandwidth Sensing:** To integrate three discrete single-axis piezoelectric transducers capable of orthogonal X, Y, and Z axis profiling. The target is to capture high-frequency components up to 15 kHz, where initial bearing degradations are most detectable [2], [7].
- **Precision Analog Conditioning:** To design and implement a custom Analog Front-End (AFE) with high-impedance buffer and amplifier stages. This objective ensures that the raw signal from the piezo sensors is accurately scaled and conditioned for the STM32 ADC without loading effects [7].
- **DMA-Driven Data Acquisition:** To develop firmware for the STM32F407VGT6 that leverages the DMA (Direct Memory Access) engine and hardware timers [6]. The goal is to achieve synchronized, jitter-free sampling across all three channels at rates between 10 kSps and 15 kSps.
- **Gapless High-Speed Logging:** To establish a robust storage pipeline using a Ping-Pong (Double) Buffering architecture and the SDIO (4-bit mode) protocol [5], [6]. This ensures that raw data is written to an SD card without any "blind spots" or data loss, fulfilling the "Black Box" forensic recording requirement.
- **Multi-Domain Fault Analysis**: To retrieve the raw data and perform advanced offline analysis using Short-Time Fourier Transform (STFT) to generate Spectrograms [9]. This objective focuses on visualizing frequency content during motor transients (startup/stop) to differentiate between imbalance, misalignment, and bearing wear [3].
- **Forensic Database Development:** To create a baseline library of vibration signatures from healthy and seeded-fault motor states, serving as a data-driven foundation for industrial diagnostic reliability.

3. Novelty

Unlike standard motor monitoring tools that rely on integrated low-frequency MEMS modules or simple amplitude tracking, this project introduces a high-performance, standalone diagnostic architecture. The novel aspects of the proposed system are:

- **High-Bandwidth Performance at Low Cost:** A significant novelty of this project is the transition from conventional MEMS accelerometers to three independent single-axis piezoelectric transducers. While commercial MEMS sensors are limited to low-frequency structural vibrations, they often miss the high-frequency "ultrasonic ringing" (10-15 kHz) that characterizes the earliest stages of bearing degradation [2], [7]. By using discrete piezoelectric disks, the system achieves the performance of professional-grade vibration analyzers at a fraction of their cost, making high-fidelity predictive maintenance accessible for a wider industrial range.
- **Engineer-Driven Sensing with Custom AFE:** Most entry-level monitoring projects use pre-assembled digital modules. This project novelly implements a custom Analog Front-End (AFE) consisting of high-impedance buffer and amplifier circuits. This design choice is critical for piezoelectric sensors to prevent signal attenuation and to match the dynamic range of the STM32 ADC. This ensures the capture of high-fidelity "micro-vibrations" that are usually lost in standard digital sensing chains [7].
- **Gapless High-Speed "Black Box" Logging via SDIO:** Handling data from three high-speed channels at 15 kSps creates a massive data flow that typical SPI or UART-based systems cannot manage without data loss. This project implements a sophisticated SDIO (4-bit mode) protocol combined with a DMA Ping-Pong buffering architecture. By bypassing the CPU during storage tasks, the system achieves a truly gapless industrial "Black Box" recording. This guarantees absolute signal integrity, a feature typically reserved for expensive, dedicated Data Acquisition (DAQ) systems [6].

- **Dynamic Visual Fingerprinting via Spectrograms:** Conventional diagnostic tools rely heavily on static FFT analysis, which assumes a stationary signal and loses all temporal information. This project introduces the use of Short-Time Fourier Transform (STFT) to generate Spectrograms [3]. This approach allows for the dynamic visualization of frequency content during motor transients, such as startup and shutdown. These "visual fingerprints" reveal transient mechanical anomalies (e.g., stator rub or load variations) that are completely invisible to standard frequency-domain analysis [4], [9].
- **Prioritization of Raw Signal over Edge Computing:** In contrast to the current trend of implementing "Edge-AI" on resource-constrained MCUs—which often introduces timing jitter and sampling errors—this project novelly prioritizes Signal Integrity. By focusing on the continuous acquisition of high-fidelity raw data, the project establishes a robust vibration signature library. This forensic approach ensures that future diagnostic models are trained on accurate, uncompromised industrial data [1], [3].

4. Proposed Method

The technical workflow of the proposed system is designed as a multi-layered "Industrial Black Box" architecture. The method focuses on capturing and preserving high-fidelity raw vibration data, followed by sophisticated multi-domain analysis. The procedure consists of four primary stages:

4.1. High-Bandwidth Sensing and Analog Front-End (AFE)

The acquisition process starts with the physical interface between the motor chassis and the processing unit.

- **Sensor Integration:** Unlike integrated MEMS sensors, three independent single-axis piezoelectric transducers are mounted orthogonally on the motor's bearing housings. This setup allows for the isolation of vibrations along the X, Y, and Z axes at frequencies up to 15 kHz [2], [7].
- **Signal Conditioning:** Piezoelectric sensors act as high-impedance sources. To prevent signal loading and ensure accuracy, a custom Analog Front-End (AFE) is implemented. This circuit features high-impedance op-amp buffers and an amplification stage to scale the signals to the 0V-3.3V range, ensuring the STM32 ADC captures the full dynamic range of the vibrations [7].

4.2. Synchronized Data Acquisition via DMA

The firmware for the STM32F407G is optimized to handle massive data throughput without disturbing the sampling accuracy.

- **Timer-Driven Sampling:** Internal hardware Timers are used to trigger three ADC channels simultaneously. This ensures that the vibration samples from all three axes are temporally aligned at sampling rates between 10 kSps and 15 kSps per channel [6].
- **DMA Ping-Pong Architecture:** To achieve gapless recording, a DMA-driven Ping-Pong (Double) Buffering logic is established. While the DMA fills "Buffer A" with fresh sensor data, the CPU processes the write operation of "Buffer B" to the storage media. This "sinusoidal" data flow prevents data overflows or gaps during high-frequency acquisition [6].

4.3. High-Speed Industrial Logging (SDIO)

Because of the high bandwidth requirement, the project utilizes the most efficient storage protocol available on the STM32 platform.

- **SDIO 4-bit Communication:** The system bypasses the low-speed SPI protocol and utilizes the SDIO (4-bit mode) interface. This provides the necessary parallel bandwidth to write continuous raw vibration blocks directly to an industrial-grade SD card in real-time [6], [10].
- **Forensic Documentation:** Data is stored in binary format with timestamp headers,

creating an industrial "Black Box" for the motor. This forensic record allows for detailed offline evaluation of the motor's mechanical state over extended periods.

4.4. Multi-Domain Offline Analysis (Spectrogram & STFT)

Retrieved raw signals are analyzed through advanced signal processing techniques to extract fault signatures.

- **Statistical Verification:** Initial screening involves calculating time-domain indicators such as Kurtosis and RMS. Kurtosis is particularly effective for piezoelectric data as it highlights the impulsive "spikes" caused by micro-cracks in bearings [1].
- **Spectrogram Generation:** The core diagnostic tool is the Short-Time Fourier Transform (STFT). By sliding a window over the raw data, the system generates Spectrograms, which represent energy distribution in both time and frequency domains [9].
- **Transient Identification:** This approach focuses on motor transients (startup/stop phases). By visualizing how frequency components shift as the motor speed changes, the system can distinguish between:
    - **Bearing Defects:** High-frequency bands and harmonic ringing [2].
    - **Rotor Imbalance:** Magnitude increases at the fundamental rotational frequency [1].
    - **Axial Misalignment:** Anomalous shifts in the 2nd and 3rd harmonics of the motor speed [3], [4].

5. Design Issues

This section outlines the technical constraints and requirements for building a high-speed "Industrial Black Box" capable of professional-grade motor diagnostics.

5.1. Project Requirements

Requirement Category	Description
Functional Requirements	Multi-channel high-bandwidth sensing using three single-axis Piezoelectric transducers; Implementation of a custom Analog Front-End (AFE) for impedance matching and signal scaling; Synchronized ADC sampling triggered by internal hardware Timers; Continuous high-speed logging via the SDIO (4-bit mode) protocol.
Usability Requirements	Industrial-grade sensor mounting for efficient vibration transfer from bearing housings; Visual status indicators (onboard LEDs) for system readiness and SD card logging activity; Portable "Black Box" form factor for quick deployment across different motor units in the field.
Performance Requirements	Minimum sampling frequency of 10-15 kHz per channel to capture high-frequency acoustic emissions; Absolute Signal Integrity ensured through DMA Ping-Pong buffering by bypassing on-chip computational load; Jitter-free multi-channel synchronization; High-speed gapless storage throughput.

5.2. Engineering Impact and Evaluation Criteria

- **Economics:** Industrial vibration analyzers and high-end DAQ systems with 15 kHz+

bandwidth can cost thousands of dollars. By utilizing affordable piezoelectric transducers and the STM32F4 Discovery board, this project provides a high-fidelity diagnostic solution at a fraction of the market price, making advanced predictive maintenance accessible for local, small-scale industries [4].

- **Environment:** Early detection of bearing cracks and shaft misalignments prevents motor burnout and catastrophic failure. This results in a reduction of industrial electronic waste and prevents the energy inefficiencies associated with operating mechanically degraded motors.
- **Sustainability:** The project promotes sustainability by extending the operational lifespan of expensive industrial assets. Through high-frequency "Black Box" logging and subsequent transient analysis via spectrograms, recurring mechanical issues are identified at the root cause, minimizing resource consumption over time [1], [3].
- **Producibility:** The hardware design relies on standard operational amplifiers and the globally supported STM32 ecosystem. This makes the system easy to mass-produce and maintain in various industrial environments without the need for proprietary, closed-source components.
- **Ethics:** The data collected is limited to mechanical vibration signatures, ensuring there are no privacy violations. The project adheres to engineering ethics by prioritizing workplace safety, aiming to protect operators from hazards like motor fires or mechanical shrapnel caused by unforeseen failures.
- **Healthcare/Safety:** Workplace safety is a critical design driver. By providing reliable early warnings, the system acts as a safety shield, allowing maintenance crews to address defects during scheduled downtimes rather than facing dangerous emergency breakdowns.
- **Security:** To ensure maximum data integrity, the system utilizes an offline storage strategy. Local SD card logging ensures that critical industrial telemetry remains physically secure and is not vulnerable to the cyber-attacks often targeted at cloud-connected industrial sensors.
- **Social and Politics:** The project supports national Industry 4.0 goals by digitalizing maintenance. It fosters local engineering expertise in high-speed data acquisition and advanced spectral analysis, reducing dependency on imported diagnostic technologies [4].

6. Project Timeline

The project is structured across two semesters, focusing on the development of a gapless data acquisition infrastructure in the first phase and advancing to multi-domain spectral fault diagnosis in the second phase.

6.1. List of Work Packages (WP)

WP No	Work Package Name	Starting – Ending Dates
WP 1	**Foundations and AFE Mastery:** Learning the STM32 hardware registers (DMA, ADC, Clock tree). Design and prototyping of the custom high-impedance Analog Piezo Front-End (AFE) circuit.	February – April
WP 2	**High-Speed Acquisition Pipeline:** Implementing synchronized 3-channel ADC sampling. Optimizing the SDIO (4-bit mode) driver for high-throughput data writing.	April – May

WP 3	**Firmware Verification & Calibration:** Developing the DMA-driven Ping-Pong (Double) Buffering logic to ensure absolute signal integrity. Establishing the standalone "Black Box" logging infrastructure.	April – May
WP 4	**Vibration Signature Data Collection:** Extensive data acquisition from healthy industrial motors and motors with "seeded" mechanical faults (damaged bearings, rotor imbalance) during the summer field period.	June – August
WP 5	**Advanced Multi-Domain Signal Analysis:** Implementing STFT (Short-Time Fourier Transform) and generating Spectrograms to visualize dynamic frequency changes during transients.	September – October
WP 6	**Diagnostic Logic & Spectral Mapping:** Developing the diagnostic identification logic based on the "Visual Fingerprints" extracted from the spectrogram datasets.	October – December
WP 7	**Validation & Graduation Thesis:** Final system-wide performance verification and preparation of the comprehensive project thesis.	December – January

6.2. List of Intermediate Outputs

Output	Related WP	Expected Date
Verified High-Impedance AFE Prototype	WP 1	Mid-April
Gapless 3-Channel Vibration Log File	WP 2	Mid-May
Complete Infrastructure Documentation (Black Box)	WP 3	End of May
Healthy vs. Faulty Vibration Database	WP 4	End of August
Spectrogram-Based Diagnostic Analysis Toolset	WP 5	End of October
Final Graduation Thesis & Presentation Materials	WP 7	January

6.3. Gantt Chart of The Project

Phase	Feb	Mar	Apr	May	Jun-Aug	Sep	Oct	Nov	Dec	Jan
WP 1	■■	■■	■■							
WP 2			■■	■■						
WP 3			■■	■■						
WP 4					■■■■					
WP 5						■■	■■			
WP 6								■■	■■	
WP 7									■■	■■

7. Risk Management

Identifying potential technical and operational bottlenecks early ensures the robustness of the system development. The following table summarizes the risks, preventative measures, and recovery strategies.

Risk	Precautions	Probability / Effect	Back Up Plan (Contingency)
Impedance Mismatch & Signal Loss: Piezo sensors have very high impedance, which can lead to significant signal attenuation before digitization.	Design and implement a custom Analog Front-End (AFE) with high-impedance op-amp buffers placed close to the sensors.	High / High	Utilize a dedicated instrumentation amplifier IC or shielded active probes to ensure signal strength for the STM32 ADC.
Data Buffer Overrun: Continuous sampling of 3 channels at 15 kHz may exceed the CPU's ability to clear buffers during high-latency SD card write cycles.	Implement a DMA-driven Ping-Pong (Double) Buffering architecture. While one buffer is being filled via ADC, the other is written to storage via SDIO.	Medium / High	Increase the internal SRAM allocation for data buffers or transition to a high-speed UHS-I rated industrial SD card.
Aliasing and Signal Clipping: High-frequency motor noise exceeding the Nyquist frequency can create ghost frequencies (aliasing) in the spectrogram.	Integrate active low-pass anti-aliasing filters within the AFE and implement adjustable gain stages to prevent ADC saturation.	High / Medium	Apply digital band-pass and notch filtering during the offline analysis stage to isolate target fault harmonics.
Industrial EMI & Supply Noise: Harsh electrical environments and 50Hz supply harmonics can mask sensitive piezoelectric vibration signals.	Use double-shielded coaxial cables for sensors and ensure complete isolation between analog and digital ground planes on the PCB.	High / Medium	Implement software-based noise reduction algorithms (e.g., adaptive filtering) during the post-processing phase.

Physical Sensor Decoupling: High-amplitude vibrations during transients may cause independent piezoelectric disks to detach or lose acoustic contact.	Employ industrial-grade mechanical clamps or specialized high-stiffness epoxy adhesives to mount sensors directly onto the bearing housings.	Low / Medium	Redesign the mounting interface using heavy-duty magnetic bases for stable yet non-invasive structural coupling.
Lack of Realistic Fault Data: Collecting signatures from actual "broken" large-scale industrial motors is often unsafe and commercially difficult.	Establish a robust "Healthy Baseline" and simulate faults locally using eccentric weights (for imbalance) or seeded-fault bearing units.	Medium / High	Supplement the custom dataset with validated open-source forensic data, such as the CWRU Bearing Data Center reference sets.

8. Conclusion

In conclusion, this project successfully introduces a high-fidelity, high-bandwidth "Industrial Black Box" system designed for the advanced condition monitoring and fault diagnosis of asynchronous induction motors using the STM32F407G-DISC1 platform [1]. By strategically integrating three independent single-axis piezoelectric transducers instead of traditional MEMS modules, the system successfully overcomes common bandwidth limitations, capturing ultrasonic acoustic emissions up to 15 kHz. This high-frequency range is critical for the early identification of incipient bearing cracks and minor mechanical anomalies that are typically missed by low-cost sensors [2], [7].

The core technical success of the project is the achievement of absolute signal integrity. By developing a custom Analog Front-End (AFE) for precision signal conditioning and implementing a hardware-level DMA-driven Ping-Pong buffering architecture, the system guarantees gapless multi-channel data acquisition [6]. Furthermore, the utilization of the SDIO (4-bit mode) protocol enables high-throughput logging directly to an SD card, fulfilling the forensic "Black Box" recording requirement essential for root-cause analysis in industrial settings [10].

From a diagnostic perspective, the transition to Short-Time Fourier Transform (STFT) based Spectrograms represents a significant analytical upgrade over conventional static FFT tools. This multi-domain approach provides a dynamic "visual fingerprint" of the motor’s health, specifically capturing transient behavior during startup and shutdown phases where faults are most visible [3], [4], [9].

The hardware and firmware infrastructure established in this phase (Spring) provides a pırıl pırıl (robust) foundation for the upcoming diagnostic stage (Fall). Ultimately, this system offers a professional-grade solution at a fraction of the cost of industrial DAQs, aligning with Industry 4.0 standards by digitizing maintenance assets, enhancing workplace safety, and complying with ISO 20816 operational vibration standards [8].

9. References

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